

Clinic Appointment No-Show Prediction

Model Evaluation, Explainability, and Fairness Reflection

Dataset: clinic_appointment_no_show_dataset.csv. The model predicts whether a patient will miss an appointment. In this assignment, no_show = 1 means the patient did not attend, while no_show = 0 means the patient attended.

Task 1: Load and Understand the Dataset

The dataset contains 380 records and 17 columns. It includes numerical variables such as patient_age, previous_no_shows, travel_distance_km, clinic_wait_time_minutes, and patient_satisfaction_score. It also includes categorical variables such as age_group, gender, region, insurance_type, appointment_type, booking_channel, and reminder_sent.

Item	Result
Dataset size	380 rows × 17 columns
Missing values	6 missing values in travel_distance_km
Duplicate records	0 duplicates
Target variable	no_show
Target distribution	0 = 228 attended; 1 = 152 no-show
Class balance	60% attended and 40% no-show; moderately balanced

Business explanation: The dataset represents clinic appointment records. The model is trying to predict whether a patient will miss a scheduled appointment. The target variable is not extremely imbalanced because both classes have enough examples for model training, although attended appointments are more common than no-shows.

Task 2: Define Features and Target Variable

The target variable is no_show. A value of 1 means the patient missed the appointment, while 0 means the patient attended. The input features are patient and appointment-related variables that may help predict no-shows.

Category	Columns
Target variable	no_show
Removed identifier	appointment_id
Fairness columns kept outside model	age_group, gender
Numerical input features	patient_age, days_before_appointment_booked, previous_no_shows, previous_appointments, appointment_hour, travel_distance_km,

	clinic_wait_time_minutes, patient_satisfaction_score
Categorical input features	region, insurance_type, appointment_type, booking_channel, reminder_sent

The identifier column is removed because it does not provide a meaningful pattern for prediction. The fairness columns age_group and gender are not used directly as model inputs; they are kept separately to evaluate whether the model behaves differently across patient groups.

Task 3: Data Preprocessing

Preprocessing is necessary because machine learning models require clean numerical input. The missing travel_distance_km values are handled using median imputation. Categorical variables are converted into numerical format using one-hot encoding. This allows the Decision Tree model to use both numerical and categorical information.

- Numerical missing values: filled with the median.
- Categorical missing values: filled with the most frequent value.
- Categorical features: converted using one-hot encoding.
- Final feature set: numerical and encoded categorical columns.

Task 4: Train/Test Split

The dataset is divided into training and testing data using an 80/20 split. The training set contains 304 records, while the testing set contains 76 records. Stratification is used so that the proportion of no-shows and attended appointments is similar in both sets.

Training data is used to teach the model patterns in the dataset. Testing data is kept separate and used only after training to evaluate how well the model performs on unseen data.

Task 5: Train a Classification Model

A Decision Tree classifier with max_depth = 4 is trained. The model learns decision rules from appointment and patient-related variables to predict whether a patient will miss an appointment. Predictions are then generated for the test set and compared with actual values.

The first few predictions should be reviewed in the comparison table produced by the Python code. Correct predictions occur when the actual value and predicted value are the same. Incorrect predictions occur when they differ.

Task 6: Evaluate the Model

Metric	Value	Meaning
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Accuracy	0.684	The model correctly predicted about 68.4% of test records.
Precision	0.667	When the model predicted no-show, it was correct about 66.7% of the time.
Recall	0.400	The model identified 40.0% of actual no-show patients.
F1-score	0.500	A balanced measure of precision and recall.

For this business problem, recall is the most important metric because the clinic wants to identify patients who are likely to miss appointments. A low recall means the model misses many true no-show patients, reducing its usefulness for proactive reminders or scheduling interventions.

Task 7: Confusion Matrix

Confusion Matrix Component	Count	Business Meaning
True Negative	40	Patients who attended and were correctly predicted as attending.
False Positive	6	Patients predicted as no-show but actually attended.
False Negative	18	Patients predicted as attending but actually missed the appointment.
True Positive	12	Patients who missed their appointment and were correctly predicted as no-show.

False positives may lead the clinic to send unnecessary reminders or take extra action for patients who would have attended anyway. False negatives are more serious because the clinic fails to identify patients who actually miss appointments. This can lead to lost appointment slots, lower clinic efficiency, and reduced access for other patients.

Task 8: Error Analysis

Error Type	Count
True Negative	40
False Negative	18
True Positive	12
False Positive	6

The most common error type is False Negative. This means the model often predicts that a patient will attend when the patient actually misses the appointment. From a business perspective, this is costly because the clinic may not take preventive action for patients who are truly at risk of no-showing.

Task 9: Cross-Validation

Fold	Accuracy
Fold 1	0.658
Fold 2	0.671
Fold 3	0.592
Fold 4	0.684
Fold 5	0.658
Mean	0.653
Standard deviation	0.032

The cross-validation scores are fairly close to each other, with a small standard deviation of about 0.032. This suggests the model is reasonably stable across different splits of the data, although the overall accuracy is moderate rather than excellent.

Task 10: Overfitting and Underfitting Analysis

The Decision Tree with `max_depth = 4` achieved training accuracy of about 0.773 and testing accuracy of about 0.684. This indicates some gap between training and testing performance, but not severe overfitting at this depth. However, when the tree depth increases beyond 6, training accuracy becomes very high while testing accuracy drops. This shows overfitting because deeper trees memorize the training data instead of generalizing well to new records.

Tree Depth	Training Accuracy	Testing Accuracy
1	0.602	0.658
2	0.701	0.632
3	0.727	0.684
4	0.773	0.684
5	0.829	0.658
6	0.859	0.684
7	0.918	0.605
8	0.957	0.592
9	0.980	0.605
10	0.990	0.605

Task 11: Feature Importance

Rank	Feature	Importance
1	travel_distance_km	0.377
2	clinic_wait_time_minutes	0.302
3	reminder_sent_Yes	0.159
4	days_before_appointment_booked	0.086
5	insurance_type_Private	0.050
6	previous_appointments	0.026

The most important feature is `travel_distance_km`, suggesting that patients who live farther from the clinic may have a higher risk of missing appointments. Clinic wait time is also important, indicating that longer wait experiences may relate to attendance behavior.

Reminder_sent is also meaningful because reminders may reduce no-shows. Feature importance does not prove cause and effect; it only shows which variables the model used most strongly for prediction.

Task 12: SHAP Explanation

SHAP explains how each feature contributes to the model prediction. In this model, the most influential SHAP features are clinic_wait_time_minutes, travel_distance_km, reminder_sent_Yes, days_before_appointment_booked, and insurance_type_Private.

For one selected test record that was predicted as no-show, travel_distance_km and clinic_wait_time_minutes increased the prediction toward no-show. Insurance_type_Private and days_before_appointment_booked pushed the prediction slightly away from no-show. SHAP is useful because it explains both global feature influence and individual prediction behavior, which basic feature importance cannot fully show.

Task 13: LIME Explanation

LIME explains one individual prediction by creating a simple local explanation around that specific record. For a selected patient record, the model prediction probability can be displayed for both classes: attended and no-show. LIME then shows which features support the predicted class and which features work against it.

For business users, LIME is helpful because it turns a complex model prediction into a more understandable explanation. For example, it may show that long travel distance, no reminder, or long clinic wait time contributed to a no-show prediction. This can help clinic staff understand why a patient was flagged and decide what intervention is appropriate.

Task 14: Fairness and Bias Reflection

Fairness was analyzed using age_group and gender. These columns were not used as model inputs but were kept to evaluate whether the model behaves differently across groups.

Group Column	Group	Records	Actual Positive Rate	Predicted Positive Rate	Accuracy
age_group	18-25	13	0.462	0.154	0.692
age_group	26-35	16	0.375	0.250	0.625
age_group	36-50	12	0.417	0.333	0.750
age_group	51+	35	0.371	0.229	0.686
gender	Female	37	0.405	0.216	0.703
gender	Male	39	0.385	0.256	0.667

The model does not perform exactly the same across all groups. Accuracy is highest for the 36-50 age group and lowest for the 26-35 age group. Gender accuracy is slightly higher for females than males. These differences do not automatically prove unfairness, but they suggest that the model should be monitored before being used in real clinic decisions.

Task 15: Final Business Interpretation

The model predicts whether a clinic patient is likely to miss an appointment. Overall, the model shows moderate performance with accuracy of 68.4%, precision of 66.7%, recall of 40.0%, and F1-score of 50.0%. The most important metric is recall because the clinic needs to identify actual no-show patients early enough to intervene.

The main error type is False Negative, meaning the model often fails to identify patients who actually miss appointments. This is important because missed no-shows can lead to wasted appointment slots and reduced clinic efficiency. The model shows some overfitting at higher tree depths, but the selected `max_depth = 4` model is a reasonable early prototype. Important features include travel distance, clinic wait time, reminder status, booking lead time, insurance type, and previous appointments. SHAP and LIME help explain why the model makes predictions, while fairness analysis shows the need to monitor performance across age and gender groups.

Final recommendation: The model is not yet ready for full real-world business use. It is suitable as an early prototype or decision-support tool, but it needs improvement before deployment. The clinic should improve recall, test additional models, tune hyperparameters, collect more data, and continue fairness monitoring.